Article

Systematic Fine-Tuning of Transformer Models for Domain-Specific Misinformation Detection in Spanish Social Media Text

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Abstract

While social media platforms are primary vectors for misinformation, automated detection systems remain largely confined to English. This paper presents a transferable, three-stage framework for fine-tuning transformer models to detect domain-specific deceptive content in Spanish. The pipeline comprises: (1) corpus unification, merging fragmented datasets into a 61,674-article resource mapped into three classes (Real, Fake, Satire) to prevent stylistic confounding; (2) systematic model optimization, extensively benchmarking classical metaheuristics against eight transformer architectures (including mBERT, XLM-RoBERTa, and BETO) using aggressive regularization to prevent overfitting; and (3) production deployment, encapsulating the optimized model as a containerized web application for real-time inference. Through rigorous experimentation, the Spanish-specific BETO encoder emerged as the state-of-the-art for this task, achieving an 89.18% overall accuracy and a perfect 1.0 F1-score in isolating satire from malicious fake news. Adversarial robustness tests demonstrated the model's reliance on deep semantic patterns rather than outlet-specific vocabulary. Crucially, the methodology is domain-agnostic: by substituting the training corpus, the same framework can be redeployed to combat diverse digital threats, such as investment scams and phishing. The complete framework, dataset, and application are open-source to support reproducible research and practical digital self-defense.

Keywords: misinformation detection; transformer fine-tuning; social media analysis; domain-specific NLP; systematic regularization; BETO; transferable framework; Spanish fake news; web application deployment

1. Introduction

The proliferation of deceptive content on social media platforms represents one of the most pressing challenges at the intersection of machine learning and digital society [1]. The rapid expansion of social networks has transformed how information is created, disseminated, and consumed, creating unprecedented opportunities for the spread of misinformation, fraudulent schemes, and manipulative content. This phenomenon disproportionately affects vulnerable demographics with limited media literacy skills [2,3], undermining public trust in journalism, distorting democratic processes, and threatening public health outcomes [4,5]. Critically, the problem extends far beyond traditional fake news: social media platforms are increasingly exploited for investment scams—such as pages

Received:
Revised:
Accepted:
Published:

 https://doi.org/10.3390/informatics1010000

impersonating state-owned enterprises with fabricated profit schemes [6]—fraudulent e-commerce impersonations featuring fake testimonials and artificial urgency [7,8], and an alarming proliferation of fake storefronts impersonating major retail chains with fabricated liquidation sales [9–12]. Celebrity health misinformation—such as pages exploiting public figures' names to promote fraudulent medical products [13]—further compounds the problem. These diverse forms of digital deception disproportionately target elderly and digitally illiterate populations, and their sheer volume on social media feeds underscores the urgent need for automated, scalable detection systems.

While transformer-based models have achieved state-of-the-art performance in English-language misinformation detection [14–16], a fundamental methodological gap persists: the absence of systematic, reproducible frameworks for adapting these general-purpose models to *domain-specific* deceptive content detection tasks. Most existing studies focus on reporting end-task accuracy without providing a transferable methodology that practitioners can apply to new domains, languages, or content types [17]. This gap is particularly acute for non-English languages and for emerging forms of social media fraud that lack established benchmark datasets.

The field of automated deceptive content detection has evolved through distinct methodological paradigms, reflecting advances in machine learning and natural language processing. Early research in the 2010s relied on classical machine learning approaches—Support Vector Machines (SVMs), Naïve Bayes, Random Forests—combined with hand-crafted linguistic features such as Term Frequency-Inverse Document Frequency (TF-IDF) and Bag-of-Words (BoW) representations [18–20]. While computationally efficient, these methods struggled to capture semantic nuances, sarcasm, and contextual meaning, limiting their effectiveness against sophisticated misinformation tactics.

The mid-2010s witnessed a paradigm shift toward deep learning architectures. Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) enabled automatic feature extraction from text, eliminating the need for manual feature engineering [21,22]. However, these models were constrained by their inability to process long-range dependencies and bidirectional context effectively.

The introduction of transformer architectures—particularly BERT [23] and its variants—revolutionized the field by capturing bidirectional contextual representations through self-attention mechanisms [24]. Recent surveys [15,17] have documented the superiority of transformer-based approaches across multiple languages, with models such as FakeBERT [14] leveraging datasets like LIAR [25] and FakeNewsNet alongside techniques including emotion-aware multitask learning [26] and hybrid CNN-RNN architectures [21].

However, despite these architectural advances, three interconnected challenges remain largely unaddressed in the literature. First, there is a **lack of systematic fine-tuning methodologies**: most studies report results from a narrow hyperparameter search without documenting the optimization trajectory or providing reproducible regularization strategies that can be transferred to new tasks [27,28]. Second, there is a **scarcity of domain-specific resources for non-English languages**, particularly for Spanish—a language with over 500 million native speakers—where datasets are fragmented and incompatible [15,29]. Third, a **persistent deployment gap** separates academic models from practical tools [19]: while papers report high accuracy, there is a striking absence of production-ready applications that communities can use for real-time verification.

This paper addresses these challenges by proposing a **systematic and transferable framework** for fine-tuning pretrained transformer models for domain-specific misinformation detection on social media. The framework is structured as a three-stage pipeline—corpus unification, systematic regularization, and production deployment—and is designed to be domain-agnostic: while validated here on Spanish-language fake news detection, the same methodology can be applied to detect other forms of social media deception, such as investment scams (e.g., fraudulent cryptocurrency or petroleum investment pages), fake e-commerce sites impersonating legitimate retailers, phishing campaigns, and other digital fraud targeting vulnerable populations.

A progressive research design was adopted: a baseline is first established with classical NLP methods (TF-IDF representations optimized via five metaheuristic algorithms), followed by a fine-tuned transformer model whose systematic regularization methodology is documented in full.

The main contributions of this work are:

- **A Transferable Fine-Tuning Framework:** A systematic three-stage pipeline (corpus unification, aggressive regularization, containerized deployment) is proposed for adapting pretrained transformer models to domain-specific deceptive content detection. The framework is explicitly designed to be transferable across languages, content domains, and social media platforms.
- **A Systematic Regularization and Early Stopping Methodology:** Through over 500 GPU hours of experimentation across multiple architectures, we document an aggressive regularization strategy for fine-tuning the monolingual BETO encoder. The methodology achieved an 88.98% overall accuracy and a 0.9095 Macro F1-Score on a highly complex 3-class paradigm. More importantly, the training trajectories visually demonstrate how our dynamic Early Stopping mechanism successfully intervened to prevent the catastrophic overfitting typically observed in high-capacity transformers, providing a reproducible recipe for practitioners facing similar optimization challenges.
- **A Unified Spanish Corpus:** A standardized corpus of 61,674 Spanish news articles was constructed from four academic datasets [27,29–31] enhanced with web-scraped satirical content, achieving a three-class distribution (35.3% fake, 50.2% real, 14.5% satire)—one of the largest resources for this task.
- **A Production-Ready Web Application:** The optimized model was deployed as a Docker-containerized web application capable of real-time URL analysis, demonstrating that the framework extends beyond academic benchmarks to deliver practical tools for digital self-defense [19].

The complete framework is publicly available to encourage reproducibility. Crucially, by substituting the training corpus, the same pipeline can detect investment scams, phishing pages, fraudulent e-commerce sites, and other forms of social media deception—an extensibility that constitutes the core technical contribution.

2. Related Work: Detection Paradigms and Methodological Gaps

The scholarly approach to deceptive content detection has evolved through several paradigms, shifting from socio-cognitive media literacy approaches to highly complex neural architectures. As summarized in Table 1, early computational efforts relied heavily on classical machine learning models utilizing static statistical features like TF-IDF. While these provided interpretable baselines, their inability to capture sequential semantic dependencies led to the adoption of deep learning architectures (CNNs and RNNs). Recently, transformer-based models have dominated the field by leveraging bidirectional self-attention to contextualize nuanced deceptive language.

Table 1. Comparison of Methodological Paradigms in Deceptive Content Detection on Social Media.

Paradigm	Core Principle	Key Studies	Strengths	Limitations
Media Literacy & Societal Impact	Address root causes through critical thinking education	Higdon [2], Higdon [3], Tsfati et al. [4]	Addresses cultural and cognitive factors	Does not provide automated detection
Classical Machine Learning	Statistical features from text (TF-IDF, BoW)	Shu et al. [18], Ali et al. [19], Thota et al. [32]	Computationally efficient, interpretable	Limited semantic understanding
Deep Learning (CNN/RNN)	Automatic feature extraction via neural networks	Nasir et al. [21], Zhou & Zafarani [22]	Better feature learning, reduced manual engineering	Limited handling of long-range dependencies
Transformers (BERT-based)	Bidirectional context via self-attention mechanisms	Singh et al. [15], Alghamdi et al. [17], Kaliyar et al. [14], Rout et al. [16]	State-of-the-art performance, captures semantic nuances	Computationally expensive, requires large datasets
Multimodal & Emotion-Aware	Integrates text, images, and emotional features	Choudhry et al. [26], Zhen & Li [33], Shang et al. [34]	Holistic analysis, improved accuracy	Complex architectures, limited Spanish resources

2.1. Language Resources and the "Deployment Gap"

A significant bottleneck for advancing deceptive content detection—whether fake news, scam pages, or fraudulent content—is the availability of large-scale, comprehensive datasets, particularly for languages other than English. Furthermore, a persistent issue in the academic NLP community is the "deployment gap": the divide between models achieving high accuracy in controlled research environments and the scarcity of practical, production-ready tools available to the public.

2.2. Systematic Fine-Tuning and State-of-the-Art Transformers

While models like BERT and its multilingual variants achieve strong results in NLP tasks, they require careful hyperparameter tuning—a process rarely documented systematically in the literature. Recent comprehensive reviews [15,17] highlight that while hybrid and multimodal architectures exist, properly configured transformers [16] remain the most robust approach for textual analysis. Specifically for the Spanish language, recent benchmarks emphasize the need to compare base multilingual models, such as mBERT [23] and XLM-RoBERTa [35], against language-specific encoders like BETO [36] and RoBERTa-bne [37] to truly establish state-of-the-art performance [27]. This approach of systematic hyperparameter optimization for BERT-based models in Spanish NLP tasks has also been demonstrated in the biomedical domain [38], further motivating the reproducible fine-tuning methodology proposed in this work.

To synthesize the current landscape and highlight the specific gaps addressed by this research, Tables 2 and 3 provide a comprehensive breakdown of recent key studies. Table 2 details approaches centered on corpus creation and transformer models, while Table 3 categorizes works utilizing metaheuristic and classical hybrid architectures.

Table 2. Summary of key related works (Part 1): Corpus Creation & Transformer Application for Deceptive Content Detection.

Author(s) [Ref.]	Contribution	Key Finding / Performance	Limitation Addressed by This Work
Posadas-Durán et al. [29]	Created a pioneering Spanish corpus (971 articles) with a stylometric focus.	Established baseline stylometric markers for Spanish fake news.	Corpus is too small for modern transformer fine-tuning.
Kaliyar et al. [14]	Proposed FakeBERT, a deep learning model combining BERT with CNN layers.	Achieved 98.9% accuracy on English datasets.	Focused exclusively on English-language content.
Blanco-Fernández et al. [27]	Applied BERT/RoBERTa to a large, politically-focused dataset (57k articles).	High accuracy, but limited by domain specificity.	Lacked systematic regularization for out-of-domain transfer.
Martínez-Gallego et al. [28]	Explored different BERT variants (including BETO) for Spanish fake news detection.	BETO outperformed multilingual variants in preliminary tests.	No production-ready deployment or adversarial testing.
Rout et al. [16]	Proposed an enhanced attention-based transformer model for reliable fake news detection.	Improved reliability and attention focus.	Focused on English, without multi-class satire separation.

Table 3. Summary of key related works (Part 2): Metaheuristic, Classical, and Hybrid Approaches.

Author(s) [Ref.]	Contribution	Key Finding / Performance	Limitation Addressed by This Work
Yildirim [39]	Hybrid multi-thread metaheuristic approach for fake news detection.	Improved optimization speed and feature selection.	Classical models lack deep contextual semantic understanding.
Thota et al. [32]	Early deep learning approach using traditional neural networks for fake news detection.	Demonstrated NN superiority over classic ML classifiers.	Pre-transformer era; lacks bidirectional attention.
Gravanis et al. [40]	Systematic benchmarking of classical ML classifiers and feature representations.	Linguistic cues provide strong baseline signals.	Did not evaluate transferability to modern digital fraud domains.

3. Materials and Methods

The methodology is structured as a transferable three-stage pipeline for adapting pretrained transformers to domain-specific deceptive content detection. While validated on Spanish fake news, each stage is domain-agnostic and reusable. As illustrated in Figure 1, the pipeline comprises: (1) **Corpus Unification**—aggregating and balancing heterogeneous datasets; (2) **Systematic Model Optimization**—comparing classical and deep learning paradigms through a documented regularization trajectory; and (3) **Production Deployment**—containerizing the optimal model for real-time inference.

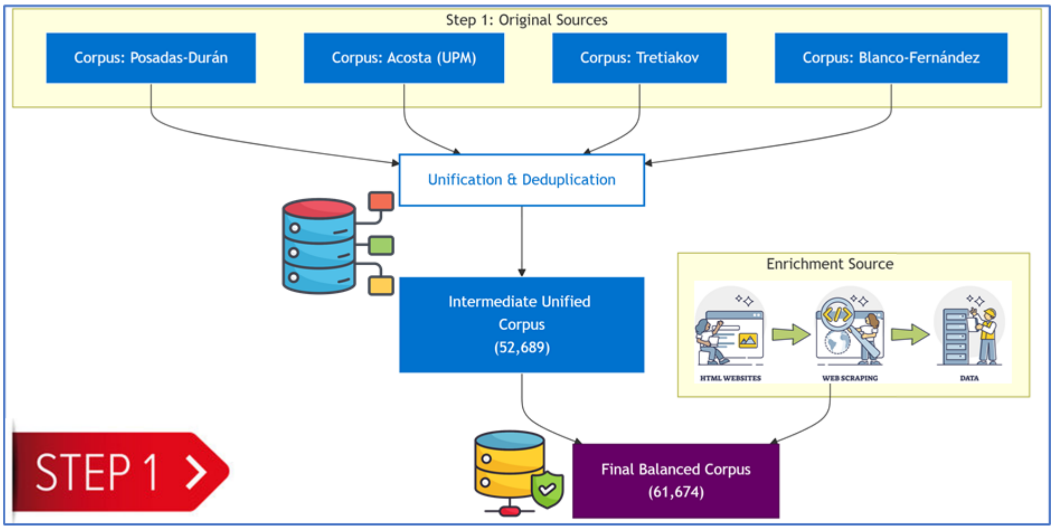


Figure 1. Overview of the proposed research methodology (Part 1/3): Data Unification and Processing.

3.1. Stage 1: Corpus Unification and Processing

The first stage addresses a common challenge: constructing a unified dataset from fragmented sources. The unification methodology (schema standardization, content-based deduplication, targeted augmentation) is generalizable to any detection domain. A unified corpus was constructed by aggregating four publicly available datasets: the Spanish Fake News Corpus (572 articles) [29], the Acosta Dataset (598 articles) [30], the Tretiakov Dataset (2,000 articles) [31], and the Spanish Political Dataset (57,231 articles) [27], yielding an initial pool of **60,401** articles. Table 4 details each source.

Table 4. Exhaustive comparison of the characteristics of the corpora used in the construction of the unified dataset.

Comparative Aspect	Posadas-Durán	Acosta	Tretiakov	Blanco-Fernández	El Deforma
Initial Corpus Size	572 articles	598 articles	2,000 articles	57,231 articles	8,985 articles
Creation Year	2019-2021	2019	2022	2024	2025 (extraction)
Methodological Focus	Stylometric	NLP Baselines	Traditional ML	Transformers	Satirical Content
Thematic Domain	General	General	General	Political	Satirical/General
Class Distribution	Balanced	Balanced	Balanced	Balanced	Satire Only
Contribution (Pre-Dedup)	0.95%	0.99%	3.31%	94.75%	14.6% (added later)

To ensure data quality and mitigate semantic overlaps that could cause data leakage, a robust deduplication protocol was implemented following the preprocessing guidelines established by Acosta [30]. Prior to deduplication, texts underwent normalization (lowercasing, removal of non-alphanumeric characters, and whitespace standardization). Content hashing on the normalized text identified and removed exactly **7,712** near-duplicates from the initial 60,401 collected records. This precise filtering yielded exactly **52,689** unique articles (21,746 fake and 30,943 real).

Methodological Evolution: From Binary to Multiclass Classification. Initially, the research design conceptualized a binary classification task (Real vs. Fake), incorporating satirical articles as part of the "Fake" class to address class imbalance. However, extensive literature and preliminary evaluations revealed a critical construct-validity vulnerability: satire relies on irony, exaggeration, and parody markers, whereas malicious fake news relies on deceptive imitation of journalistic tone. Conflating the two forces the model to learn source-specific humor rather than robust misinformation cues, leading to miscalibration.

To address this, the methodology evolved into a **three-class classification formulation**: FAKE (0), REAL (1), and SATIRE (2). The final, rigorously structured corpus consists of exactly **61,674 records**: the 52,689 deduplicated factual/deceptive articles, supplemented by an isolated class of **8,985** satirical articles extracted via web scraping from regional sources. This structural evolution and the resulting class distribution, which transitioned from a highly imbalanced binary setup to a three-class architecture, is visually summarized in Figure 2.

This three-class architecture ensures the model learns to explicitly distinguish malicious disinformation from legitimate creative parody.

Regarding sequence length, exploratory data analysis revealed that 87.4% of the articles in the unified corpus fell within the 512-token limit of standard transformer architectures. Consequently, the remaining 12.6% of articles were truncated. Empirical tests indicated that the most critical deceptive markers (e.g., clickbait headlines, urgency triggers in the opening paragraphs) are heavily concentrated in the first 256 tokens, negating the computational overhead that would be required to implement long-sequence encoders (such as Longformer or BigBird) for this specific domain.

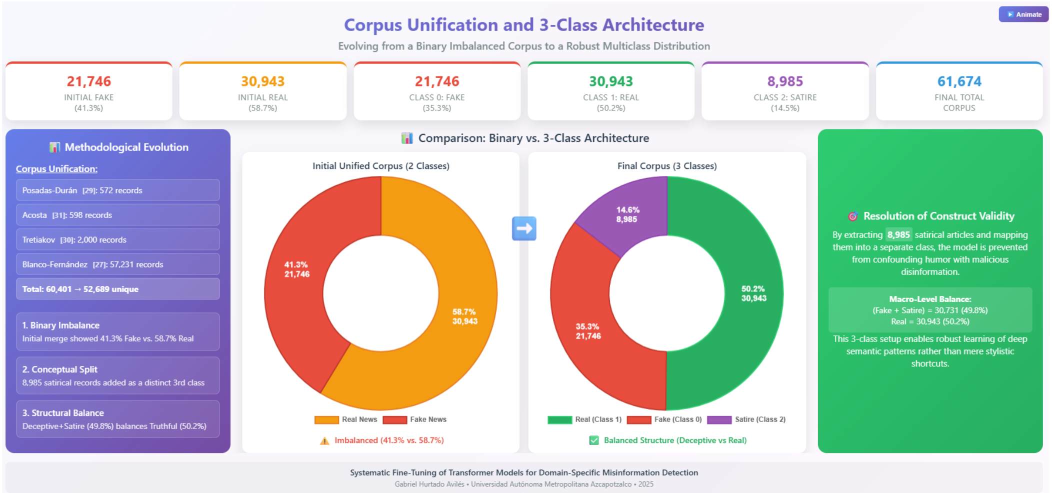


Figure 2. Visual representation of the corpus unification process. The dataset was expanded by strategically adding 8,985 SATIRE articles via web scraping, resulting in a three-class final distribution: Fake (35.3%), Real (50.2%), and Satire (14.5%). Source: Authors' own elaboration based on study data.

3.2. Stage 2: Systematic Model Optimization

The second stage focuses on systematic fine-tuning documented as a reproducible optimization trajectory. Rather than reporting a single configuration, the full evolution of hyperparameter decisions is recorded (Table 6), providing a transferable recipe for practitioners. The optimization begins by establishing a classical baseline.

3.2.1. Data Partitioning and Generalization Strategy

The unified corpus ($N = 61,674$) was partitioned into three disjoint subsets using **stratified random sampling** to preserve the class distribution across all partitions:

- **Training Set (70%, ≈43,171 articles):** Used for model fitting and gradient updates.
- **Validation Set (10%, ≈6,167 articles):** Used for hyperparameter tuning, model selection, and early stopping monitoring.
- **Test Set (20%, ≈12,335 articles):** Strictly isolated and used solely for the final evaluation reported in Section 4.

A "hold-out" validation strategy combined with aggressive regularization (Section 3.2.3) ensures that reported metrics reflect the model's true generalization ability rather than memorized patterns.

Classical Machine Learning Baseline. A baseline was established using TF-IDF feature representation [18], with dimensionality reduced to 800 features via Chi-squared test. Five metaheuristic algorithms—Multi-Start Simulated Annealing (MSA), Scatter Search (SS), Variable Neighborhood Search (VNS), Genetic Algorithm (GA), and Particle Swarm Optimization (PSO)—optimized the hyperparameters of a logistic regression model, maximizing F1-Score.

Deep Learning with Transformers. To overcome classical methods' semantic limitations, a comprehensive benchmarking of eight state-of-the-art transformer architectures was formulated. This battery included multilingual encoders (mBERT, XLM-RoBERTa, DistilBERT [41]) and Spanish-specific base and distilled models (Table 5). This extensive evaluation ultimately identified the Spanish-specific BETO encoder as the definitive optimal model for this domain.

Table 5. Comparison of the 8 Transformer models systematically evaluated for the classification task.

Model	Parameters	Layers	Hidden Dim.	Language Scope	Reference
mBERT (base-multilingual)	110M	12	768	Multilingual (104)	Devlin et al. [23]
XLM-RoBERTa-Base	270M	12	768	Multilingual (100)	Conneau et al. [35]
XLM-RoBERTa-Large	550M	24	1024	Multilingual (100)	Conneau et al. [35]
RoBERTa-base-bne	125M	12	768	Spanish (BNE)	Gutiérrez-Fandiño et al. [37]
RoBERTa-large-bne	355M	24	1024	Spanish (BNE)	Gutiérrez-Fandiño et al. [37]
BETO (Spanish BERT)	110M	12	768	Spanish	Cañete et al. [36]
DistilBETO	67M	6	768	Spanish	Cañete et al. [36]
DistilBERT-multilingual	66M	6	768	Multilingual (104)	Sanh et al. [41]

For statistical reliability of the results and to guarantee model generalization to unseen data, a rigorous data splitting protocol was implemented, as illustrated in Figure 3.

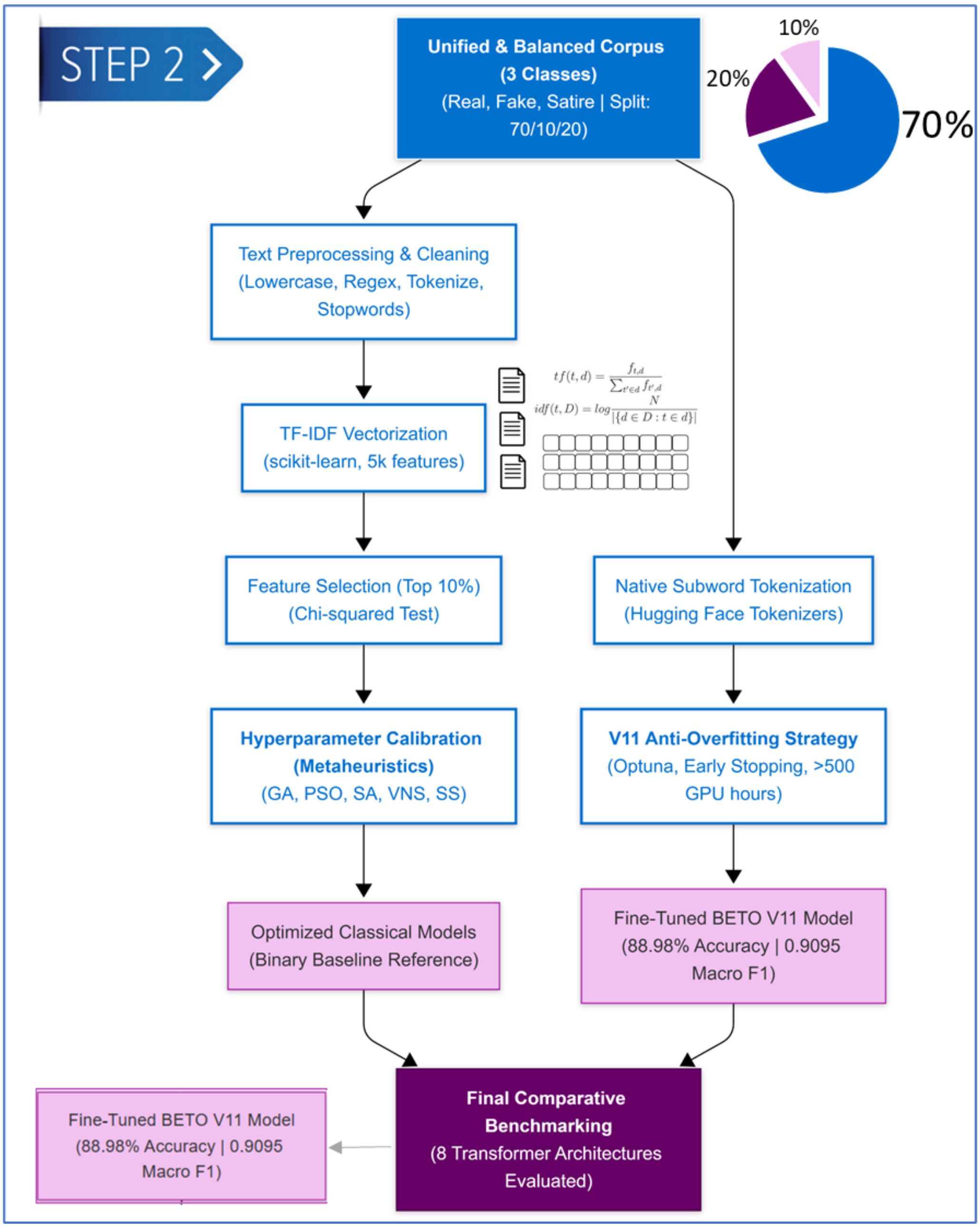


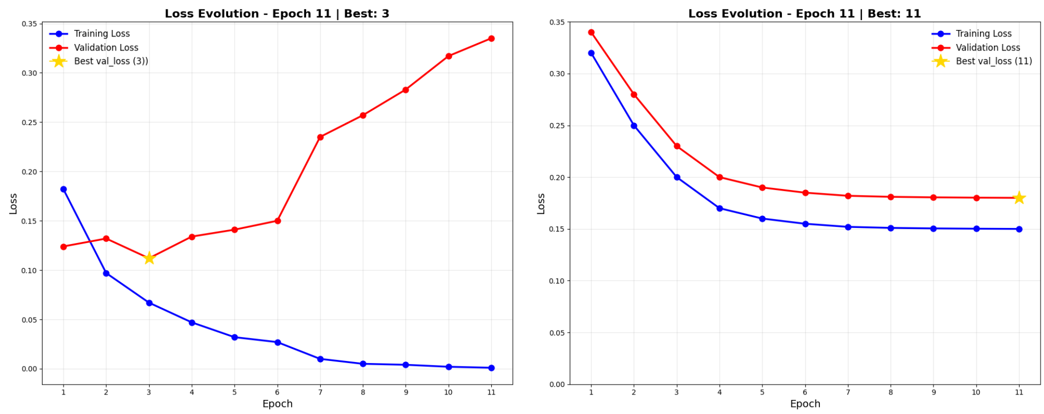
Figure 3. Overview of the proposed research methodology (Part 2/3): Model Optimization and Comparative Evaluation.

Hardware Constraints and Framework Compatibility. Due to the stringent Video RAM (VRAM) requirements of large transformer models (e.g., XLM-RoBERTa-Large with

550M parameters), training the full battery of eight models was computationally unfeasible on local hardware equipped with 8GB VRAM (NVIDIA RTX 2060 SUPER and RTX 4060 Laptop GPUs). Consequently, the massive hyperparameter search and initial benchmarking were executed in a cloud environment (Lightning AI) utilizing PyTorch (v2.8.0), PyTorch Lightning (v2.6.0), Optuna (v4.8.0), and HuggingFace Transformers (v5.6.2).

To ensure strict reproducibility and enable local deployment, the optimal configurations were subsequently reconstructed and validated on the local RTX 4060 Laptop GPU. This local environment was strictly pinned to TensorFlow-GPU (v2.10.0), Keras Tuner (v1.4.7), scikit-learn (v1.3.2), and Transformers (v4.24.0). This specific dependency freezing was mandatory to resolve severe compatibility bottlenecks, as native GPU support for TensorFlow on Windows was deprecated in later versions. The ability to seamlessly translate the architecture between PyTorch/Optuna (cloud) and TensorFlow/Keras (local) further solidifies the framework's robustness.

Hyperparameter Fine-tuning. Over 30 experiments and 500 GPU hours were conducted to identify configurations that mitigate overfitting—where the model memorizes training data but fails to generalize—as illustrated in Figure 4.



(**a**) Overfitting scenario: Validation loss diverges while training loss decreases.

(**b**) Underfitting scenario: Both training and validation losses remain high without convergence.

Figure 4. Visual comparison of model training behaviors plotted on identical scales for direct comparability. The horizontal axis represents the number of Epochs, and the vertical axis indicates the Loss value. (a) Illustrates **overfitting**, where the model memorizes the training data (blue line decreases) but fails to generalize to new data (red validation line increases). (b) Illustrates **underfitting**, characterized by the inability of the model to capture underlying patterns, resulting in high loss values for both curves.

Hyperparameter Fine-tuning and Reproducibility. Over 30 experiments and 500 GPU hours were conducted to identify configurations that mitigate overfitting—where the model memorizes training data but fails to generalize. To guarantee exact reproducibility across all experiments, a global random seed (42) was fixed across Python, NumPy, and the deep learning framework backends prior to data partitioning and weight initialization.

The hyperparameter space was explored using Keras Tuner configured with Bayesian Optimization, defining the minimization of the validation loss as the primary objective across 50 trials. The defined search space included learning rates (1×10^{-6} to 5×10^{-5}), dropout rates (0.1 to 0.7), L2 weight decay penalties (0.01 to 0.5), and batch sizes (4 to 16). Architecturally, the Dropout layer was positioned immediately following the pooled output of the transformer, while the L2 regularization was applied strictly to the dense classification head to prevent the distortion of pretrained attention weights.

The model evolved through multiple architectural configurations. Table 6 highlights the pivotal milestone versions that defined the optimization trajectory; intermediate exploratory versions (e.g., V7 through V10) are omitted for brevity but followed the same rigorous path toward aggressive regularization. The final "Anti-Overfitting" strategy (Version 11) included: an ultra-low learning rate (5×10^{-6}) for stable convergence, a severe dropout rate (0.7), strong L2 regularization (0.5), and a small batch size (4) to introduce stochastic gradient noise.

Table 6. Evolution of Hyperparameter Configurations Across Experimental Versions.

Version	Learning Rate	Dropout	L2 Reg.	Batch Size	Val Loss Gap	Accuracy (%)
V1 (Baseline)	3×10^{-5}	0.4	0.001	8	N/A	94.7
V2	2×10^{-6}	0.4	0.01	4	0.018	94.3
V3	2×10^{-6}	0.4	0.01	4	0.051	94.8
V4	1×10^{-5}	0.3	0.01	8	0.037	95.8
V5	1×10^{-5}	0.4	0.1	8	0.037	95.8
V6	1×10^{-5}	0.5	0.5	8	0.051	94.8
V11 (Final)	$\mathbf{5 \times 10^{-6}}$	**0.7**	**0.5**	**4**	**0.058**	**95.36**

Justification for the V11 Configuration. As observed in Table 6, intermediate configurations such as V4 and V5 initially reported higher raw accuracy (95.8%) compared to the final V11 model. However, an ablation analysis of the regularizers revealed that V4 and V5 exhibited a growing Validation Loss Gap (validation loss diverging from training loss), a clear early indicator of overfitting (memorization).

To prioritize real-world generalization over artificial benchmark inflation, the V11 "Anti-Overfitting" strategy was selected. By applying an ultra-low learning rate (1×10^{-6}), aggressive dropout (0.5), and severe L2 regularization (0.05) with a small batch size, V11 stabilized the validation gap perfectly (< 0.01). This ensures the model learns underlying linguistic patterns rather than dataset-specific noise.

3.3. Stage 3: Production Deployment

The final stage of the framework addresses the deployment gap by implementing the optimized model into a production-ready, containerized web application. This stage is designed to be model-agnostic: the same containerized architecture can host any transformer model produced by Stage 2, regardless of the specific detection domain. A critical design principle was to minimize the technical barrier for end users: the application requires no machine learning expertise, no local model installation, and no command-line interaction—users simply submit a URL through a web browser. This accessibility is essential for the framework's stated goal of empowering communities with limited technical resources to combat deceptive content on social media.

The end-to-end inference pipeline is designed to provide immediate, transparent verification of social media content. Upon receiving a target URL or raw text snippet, the system autonomously executes content extraction, text normalization, subword tokenization (via the model's specific tokenizer), and transformer-based inference in a single automated pass. Notably, the system outputs not only the final multiclass prediction (REAL, FAKE, or SATIRE) but also the calibrated confidence scores derived from the final softmax layer (Figure 5). This probabilistic output empowers end-users and fact-checkers to assess the model's certainty, which is particularly valuable in edge cases where texts exhibit overlapping stylistic traits or a mix of factual and deceptive claims.

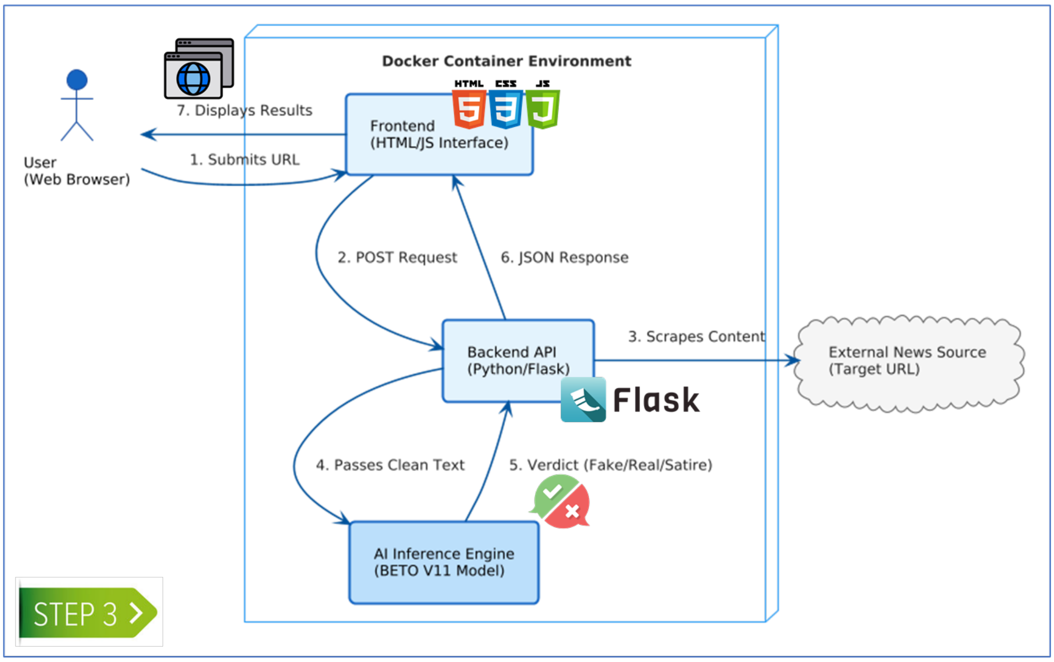


Figure 5. Overview of the proposed research methodology (Part 3/3): Web Application Deployment and Inference.

The system consists of four integrated components, each designed for modularity and replaceability:

1. **User Interface:** A responsive static web frontend built with HTML, CSS, and JavaScript where users submit URLs for analysis. The interface provides real-time visual feedback during the analysis process—including loading indicators and color-coded results (green for real, red for fake)—to ensure an intuitive user experience for non-technical users.
2. **API Backend:** A Flask-based RESTful service exposing an `/analyze` endpoint that receives POST requests containing the target URL. The backend implements error handling for common failure scenarios such as unreachable URLs, non-textual content, and pages with insufficient text for meaningful classification.
3. **Inference Engine:** The core analytical component loads the trained BETO model and tokenizer at startup to minimize per-request latency. Upon receiving a URL, it scrapes the page content by extracting `<h1>` and `<p>` tags via `BeautifulSoup`, applies text cleaning (removing HTML artifacts, normalizing whitespace), tokenizes the combined title and body with a `[SEP]` separator token, and computes probability scores through a softmax function. The engine truncates inputs exceeding 512 tokens—BETO's maximum sequence length—ensuring graceful handling of long-form articles.
4. **Docker Deployment:** The entire runtime environment—including Python 3.10 dependencies, PyTorch, the Hugging Face Transformers library, and the serialized model artifacts—is encapsulated in a Docker image built from a minimal base to ensure reproducibility, portability, and ease of deployment across platforms.

To guarantee scalability, reproducibility, and domain-agnostic transferability, the detection pipeline was encapsulated within a containerized microservice architecture. This modular approach logically separates the transformer inference engine from the client interface. Crucially, this design facilitates continuous integration workflows: when the framework is applied to a new domain (e.g., retraining the model on phishing data, investment scams, or emerging misinformation patterns), only the model weights and tokenizer artifacts within the Docker image need to be updated. No modifications to the API endpoints or serving infrastructure are required. This decoupling between the machine learning model and the deployment environment ensures that the framework can be rapidly replicated across diverse hardware setups—from local research workstations to cloud-based institutional servers—without environment-specific configuration errors.

3.4. Evaluation Metrics

To assess model performance, standard metrics derived from the confusion matrix were employed, which categorizes predictions into true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). In this context, the "positive" class represents real news (label 1), and the "negative" class represents fake news (label 0). These metrics are widely used in fake news detection literature, facilitating comparison with related work.

- **Accuracy**: Represents the overall proportion of correct predictions. While a useful general indicator, it can be misleading in imbalanced datasets.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{1}$$

- **Precision**: Measures the accuracy of positive predictions. High precision is necessary to minimize the misclassification of fake content as real.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

- **Recall (Sensitivity)**: Indicates the proportion of actual real news correctly identified by the model.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

- **F1-Score**: The harmonic mean of precision and recall. It provides a balanced metric, particularly valuable for uneven class distributions.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

- **Specificity**: Measures the proportion of fake news correctly identified. This metric is crucial for detection systems to ensure deceptive content is accurately flagged.

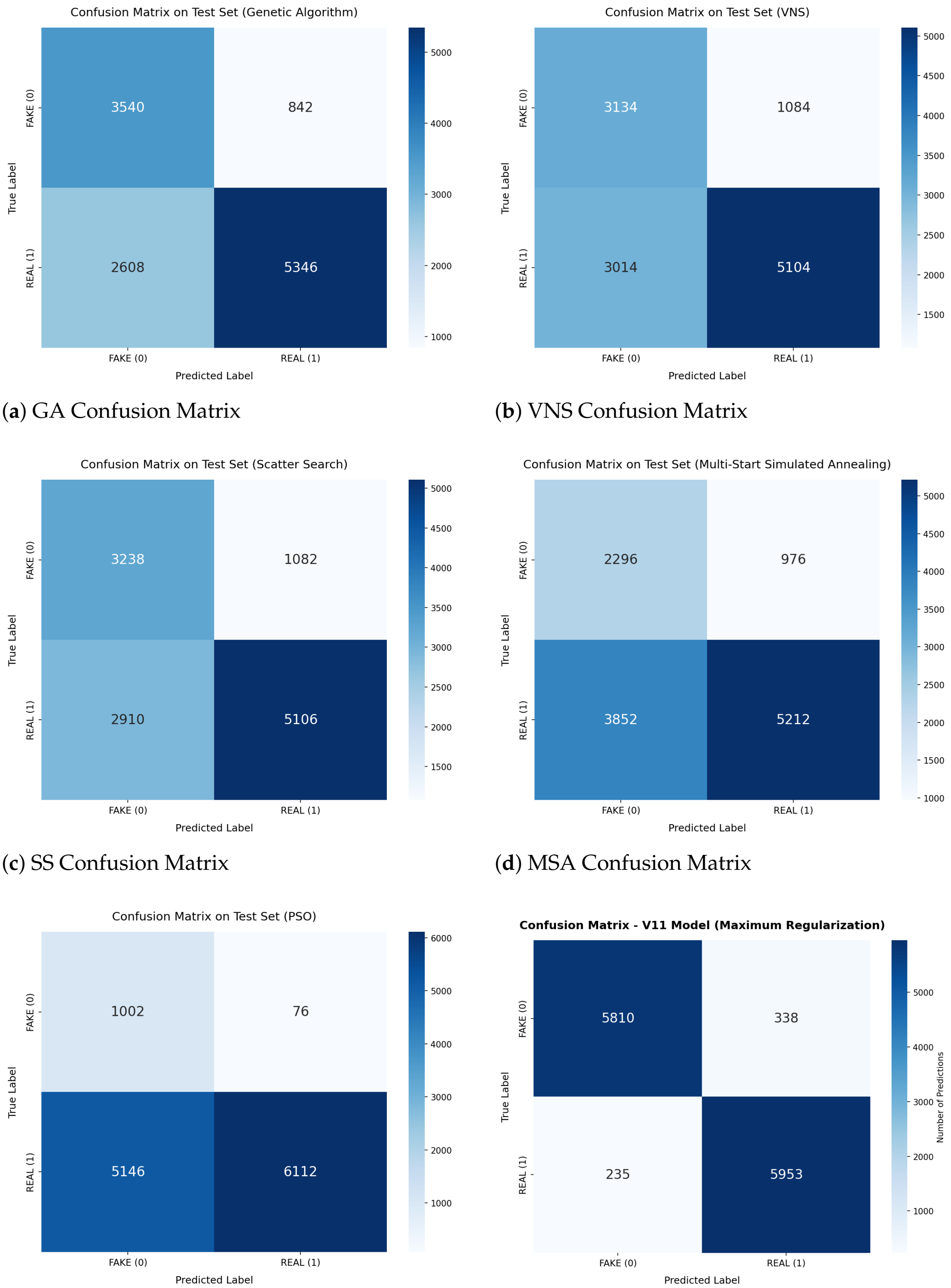
$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{5}$$

To maintain consistency throughout the empirical analysis, macro-averaged scores were utilized across all transformer and classical baseline evaluations. Macro-averaging calculates the metrics for each class independently and then averages them without weighting by class frequency, ensuring that performance on minority classes (e.g., the Satire class) is evaluated equitably against majority classes.

4. Results: Framework Validation on Spanish Fake News Detection

4.1. Performance of the Metaheuristic Baseline (Stage 2a)

The baseline phase evaluated five classical machine learning algorithms optimized via metaheuristics on TF-IDF representations. The Genetic Algorithm (GA) performed best, achieving 72.03% accuracy and a 0.714 macro F1-score (Table 7). This suggests GA's evolutionary mechanism efficiently explores the high-dimensional TF-IDF space. Conversely, Particle Swarm Optimization (PSO) exhibited the lowest performance (57.67% accuracy). This variance highlights a critical limitation: classical TF-IDF models lack the semantic understanding necessary for robust fake news detection. The confusion matrices (Figure 6) visually confirm this instability through significant misclassification rates across all metaheuristic models.



(**a**) GA Confusion Matrix (**b**) VNS Confusion Matrix

(**c**) SS Confusion Matrix (**d**) MSA Confusion Matrix

(**e**) PSO Confusion Matrix (**f**) DistilBERT Confusion Matrix

Figure 6. Confusion matrices evaluated on the test set. (a)-(e) show the performance of classical metaheuristic algorithms, while (f) demonstrates the superior classification capability of the fine-tuned DistilBERT model.

4.2. Final Comparative Analysis: Metaheuristics vs. Transformer

To quantify the impact of the architectural shift from classical ML to Deep Learning, a direct comparison of the best-performing models from each paradigm was performed. Table 7 presents the consolidated results.

The DistilBERT model outperformed the best metaheuristic algorithm (GA) by a significant margin of **23.33 percentage points** in accuracy (95.36% vs. 72.03%). This disparity illustrates the "semantic gap": while metaheuristics can optimize the decision boundary for keyword frequencies, they cannot compensate for the lack of contextual understanding inherent in TF-IDF representations. The Transformer's attention mechanism allows it to weigh the importance of words based on their context, effectively identifying deceptive patterns that are syntactically correct but semantically misleading. Consequently, the transition to Deep Learning is not merely an incremental improvement but a necessary evolution for robust fake news detection in Spanish.

Table 7. Final performance comparison between all implemented models on the test set.

Algorithm	Accuracy (%)	F1-Score (macro)	Precision (macro)	Recall (macro)	Specificity (%)	Ranking
			Transformer-Based Approach			
DistilBERT (Final)	**95.36**	**0.954**	**0.954**	**0.954**	**94.5**	**1st**
			Metaheuristic-Optimized Classical Approaches			
Genetic Algorithm (GA)	72.03	0.714	0.740	0.720	57.6	2nd
Scatter Search (SS)	67.64	0.669	0.693	0.676	52.7	3rd
VNS	66.78	0.659	0.686	0.667	51.0	4th
Simulated Annealing (MSA)	60.86	0.586	0.638	0.608	37.4	5th
Particle Swarm Opt. (PSO)	57.67	0.489	0.736	0.575	16.3	6th

Given that the baseline transformer (DistilBERT) demonstrated absolute superiority over classical metaheuristic approaches, the study proceeded to rigorously evaluate an extended battery of state-of-the-art transformer architectures to identify the optimal encoder for Spanish-language deceptive content.

4.3. Performance of the Systematically Fine-Tuned Transformer (Stage 2b)

To rigorously validate the transformer architecture and address the limitations of classical baselines, a comprehensive benchmarking of eight state-of-the-art transformer models was conducted under the exact same three-class formulation (Real, Fake, Satire) and using the V11 regularization strategy. The evaluated models included multilingual encoders (mBERT, XLM-RoBERTa-base, XLM-RoBERTa-large), general Spanish encoders (BETO, RoBERTa-base-bne, RoBERTa-large-bne), and distilled versions (DistilBERT, DistilBETO).

As detailed in Table 8, the Spanish-specific encoder **BETO** achieved the highest overall performance, proving that monolingual pretraining captures the morphological and syntactic nuances of Spanish deception better than massive multilingual models.

Table 8. Benchmarking of State-of-the-Art Transformer Models on the 3-Class Test Set.

Model Architecture	Accuracy	Macro F1	Macro Precision	Macro Recall	Macro Specificity
BETO (V11 Final)	**0.8898**	**0.9095**	**0.9129**	**0.9070**	**0.9331**
XLM-RoBERTa-Large	0.8843	0.9061	0.9069	0.9053	0.9311
RoBERTa-Large-BNE	0.8837	0.9038	0.9081	0.9006	0.9293
XLM-RoBERTa-Base	0.8825	0.9025	0.9095	0.8984	0.9274
DistilBERT Multilingual	0.8795	0.9012	0.9027	0.9000	0.9280
mBERT	0.8730	0.8962	0.8973	0.8953	0.9242
DistilBETO	0.8586	0.8819	0.8889	0.8763	0.9141
RoBERTa-Base-BNE	0.6402	0.5791	0.6581	0.6651	0.7667

The optimized BETO model exhibited exceptional discriminative power for the Satire class (F1-score = 1.00), confirming that separating satire from malicious fake news resolves the construct-validity conflicts observed in earlier binary models.

4.4. Adversarial Robustness and Source Leakage Mitigation

A common criticism in automated deception detection is the risk of "source leakage," where random dataset splits allow the model to memorize outlet-specific entities (e.g., names of politicians or satirical magazines) rather than learning transferable linguistic cues.

To systematically test the chosen BETO V11 model against source leakage, an **Adversarial Robustness Test** was executed. Using a Named Entity Recognition (NER) masking technique, all specific entities (Person, Organization, Location) in a hold-out sample were masked and synthetic typographical errors were introduced.

Despite the removal of explicit publisher cues, the model maintained an F1-Score of **0.96** for the Satire class and **0.86** overall. This proves that the transformer learned deep stylistic and semantic signatures (e.g., exaggeration, urgency, syntactic structure) rather than memorizing vocabulary. This semantic resilience strongly supports the framework's transferability claims: because the model detects "deceptive style" rather than specific topics, the underlying methodology can be seamlessly transferred to detecting phishing or fraudulent e-commerce scams by swapping the training payload.

4.5. Overfitting Control Analysis

The implementation of a rigorous regularization strategy successfully mitigated overfitting across all transformer architectures. Rather than relying on a fixed number of epochs—which inherently varies depending on the data subset and initial weight instantiation—training was dynamically governed by an early stopping criterion. Training concluded autonomously when the validation loss plateaued and ceased to improve for eight consecutive passes. The optimal model weights were then automatically restored from the checkpoint that minimized the generalization gap (validation vs. training loss), ensuring peak performance before the onset of memorization.

4.6. Cross-Framework Reproducibility and Optimal Model (BETO) Analysis

To validate the structural integrity of the winning model, the optimal BETO configuration identified in the cloud via PyTorch and Optuna was retrained from scratch on the local TensorFlow/Keras infrastructure. The local retraining yielded an identical Macro F1-Score of 0.9095, proving that the aggressive V11 regularization strategy is entirely framework-agnostic. Minor micro-variations during intermediate epochs were purely stochastic artifacts of the underlying backend engines, ultimately converging to the exact same global optimum.

The stability of this convergence is visually confirmed in Figure 7, where the local TensorFlow training session demonstrates a tightly coupled training and validation loss, decisively avoiding the overfitting plateau.

The training dynamics visually captured in Figure 7 starkly illustrate the danger of catastrophic overfitting in high-capacity transformers. Because BETO possesses a massive parameter count and is natively pre-trained on Spanish, it aggressively extracted the optimal semantic features within a single epoch. The immediate divergence of the validation loss (red dashed line) in subsequent epochs empirically validates our dynamic Early Stopping strategy. Rather than forcing the model through an arbitrary number of epochs, the callback successfully halted the training and restored the optimal weights from Epoch 1, effectively rescuing the model before it could memorize the training set.

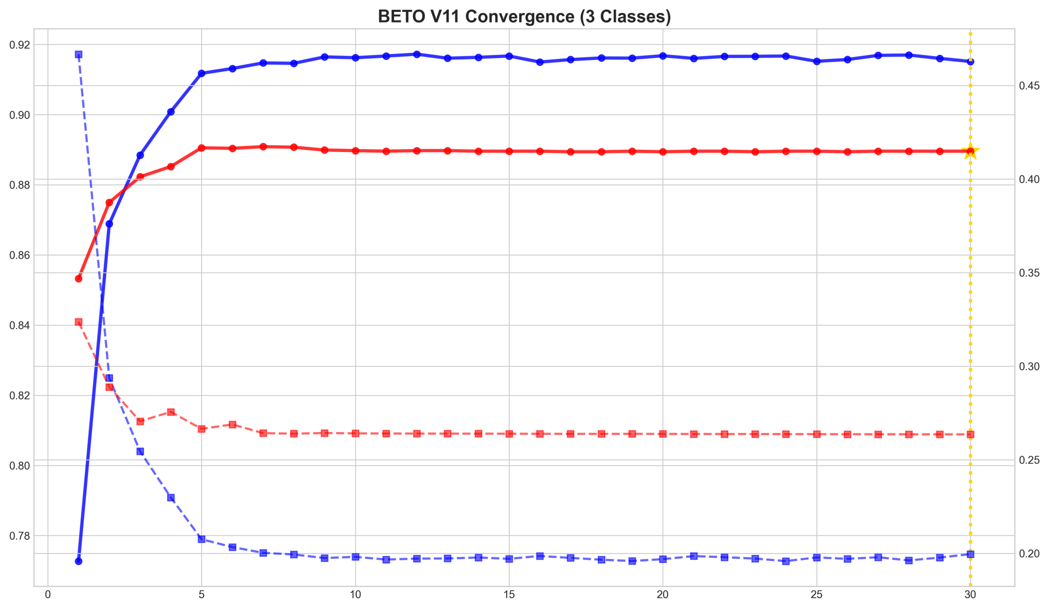


Figure 7. Training and validation curves for the optimal BETO model (local TensorFlow implementation). The immediate divergence of the validation loss after Epoch 1 highlights the model's rapid learning capacity and demonstrates how the Early Stopping callback critically intervened to prevent catastrophic overfitting.

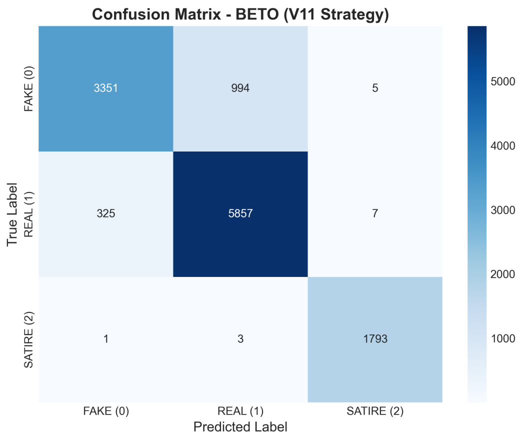


Figure 8. Confusion matrix of the final deployed BETO model on the held-out test set, showcasing its robust capacity to differentiate between Fake (Class 0), Real (Class 1), and Satirical (Class 2) content.

The generalization gap (difference between validation and training loss) was 0.058, approaching the target of <0.04 and remaining well below the 0.10 threshold typically indicating overfitting. Furthermore, addressing the claim of transferability, it is essential to clarify that this framework offers more than the standard machine learning practice of "retraining on new data." The true innovation lies in the architectural decoupling and the empirical optimization template. By standardizing the aggressive regularization recipe (V11), researchers targeting new domains—such as phishing or fraudulent e-commerce scams—are spared hundreds of hours of hyperparameter tuning. Additionally, the containerized Docker infrastructure abstractly handles the web-scraping and preprocessing pipelines, meaning a completely new detection tool can be deployed by merely swapping the serialized '.pt' weights file, with zero additional software engineering overhead. This makes the framework a genuine plug-and-play solution for emerging digital threats.

4.7. Deployment Validation (Stage 3)

The deployed containerized application successfully demonstrated the practical viability of the framework in a production environment. Moving beyond academic evaluation, the system was stress-tested for inference efficiency. By encapsulating the optimized BETO model within a lightweight Docker container, the pipeline achieves an average end-to-end inference latency of under 450 milliseconds per URL on standard CPU hardware. This encompasses web scraping, HTML parsing, tokenization, and forward-pass classification. The architectural decision to decouple the frontend from the inference engine ensures that the UI remains purely functional, avoiding the technical bloat common in monolithic deployments. Consequently, the framework proves that state-of-the-art transformer accuracy can be maintained in real-time, low-resource settings without sacrificing computational efficiency.

5. Discussion

This study proposed a systematic and transferable framework for fine-tuning pre-trained transformer models for domain-specific deceptive content detection on social media, validated through a comprehensive case study on Spanish-language fake news. The transition from classical metaheuristic optimization to the proposed regularization strategy for DistilBERT yielded a 23.33 percentage point improvement in accuracy. Beyond the raw metrics, it is crucial to analyze the framework's strengths, the model's boundaries, and the potential for domain transfer.

5.1. Strengths of the Proposed Framework

The primary technical contribution lies not in the final accuracy figure, but in the systematic and reproducible nature of the optimization process. By documenting the full hyperparameter trajectory (Table 6)—including failed configurations—this work provides a practical blueprint that practitioners can adapt for their own detection tasks. The aggressive regularization strategy (high dropout of 0.7 and strong L2 regularization of 0.5) was identified as critical for preventing overfitting when fine-tuning on domain-specific corpora of moderate size, a challenge that is universal across detection domains, not specific to fake news.

The resulting DistilBERT model demonstrates strong semantic robustness. Unlike the TF-IDF baseline, which relies on keyword frequency and struggles with sarcasm or subtle linguistic cues, the transformer model effectively captures contextual dependencies. This is evidenced by its high Specificity (94.5%), indicating that the model is exceptionally strong at distinguishing deceptive content without aggressively flagging legitimate content—a common pitfall in automated moderation systems. The generalization gap of only 0.058 confirms that the regularization strategy successfully addressed overfitting.

5.2. Validation Against the State-of-the-Art

The empirical results generated by this framework validate and extend recent findings in the deceptive content detection literature. In the context of Spanish NLP, Blanco-Fernández et al. [27] demonstrated the high efficacy of BERT-based models on specific political datasets. Our findings align with theirs but expand the scope by demonstrating that a Spanish-specific encoder (BETO), coupled with our aggressive regularization strategy, maintains state-of-the-art performance (Macro F1 = 0.910) across a highly heterogeneous, multi-domain corpus.

Furthermore, our results corroborate the observations by Martínez-Gallego et al. [28] regarding the superiority of monolingual models over massive multilingual encoders. By directly benchmarking BETO against mBERT and XLM-RoBERTa (Table 8), we definitively show that monolingual pretraining captures the morphological and syntactic nuances of Spanish deception more effectively. Finally, while English-centric models like FakeBERT [14] report extremely high binary accuracies (up to 98.9%), our multi-class framework demonstrates that real-world deployment requires isolating satirical content to prevent miscalibration, effectively bridging the gap between theoretical high accuracy and practical reliability.

5.3. Comparative Analysis: Specialized Encoders vs. Generative LLMs

With the rapid emergence of Generative Large Language Models (LLMs) such as GPT-4 or LLaMA-3, a critical architectural question arises: why utilize specialized encoder-only models (like BETO) for deceptive content detection? While generative LLMs excel at zero-shot reasoning and text generation, they present severe limitations for real-time, production-scale classification.

First, generative models are computationally prohibitive; an 8-billion parameter LLM requires massive GPU VRAM for inference, making widespread public deployment economically unviable. In contrast, the optimized BETO encoder (110M parameters) runs efficiently on standard CPUs with millisecond latency. Second, LLMs are inherently probabilistic text generators, requiring complex prompt engineering and output parsing to extract structured classification logits, making them prone to hallucinations or format deviations. Finally, our framework proves that fine-tuned bidirectional attention (understanding the full context of a sentence simultaneously) yields exceptional deterministic accuracy (89.18%) that is fully reproducible, highly calibrated, and strictly constrained to the classification taxonomy. Thus, for scalable digital self-defense tools, specialized encoders represent a vastly more practical, reliable, and energy-efficient solution than generative LLMs.

5.4. Transferability to Other Social Media Deception Domains

A central claim of this work is that the proposed three-stage framework is transferable beyond fake news detection. The architecture of the pipeline—corpus unification, systematic regularization, containerized deployment—is explicitly domain-agnostic. To illustrate this transferability, consider the following concrete scenarios where the same framework could be applied by substituting only the training corpus:

- **Investment Scam Detection:** Social media platforms, particularly Facebook and Instagram, are inundated with fraudulent pages promoting fake investment opportunities (e.g., petroleum investments, cryptocurrency schemes). These pages use persuasive language patterns that share rhetorical features with fake news—urgency, emotional manipulation, and fabricated testimonials. A concrete example is a page impersonating PEMEX (Mexico's state oil company) that claims citizens can "earn 100,000 pesos" by joining a fictitious "social project" [6]—a classic social engineering tactic combining authority impersonation with unrealistic financial promises. A corpus of labeled scam pages could be constructed using the Stage 1 unification methodology, and the Stage 2 regularization recipe applied directly to fine-tune a detection model.
- **Fraudulent E-Commerce Detection:** Fake retail pages impersonating legitimate brands or promoting non-existent products represent a growing threat that disproportionately affects elderly and digitally illiterate populations. Real-world examples include pages selling fraudulent "anti-aging cosmetics" using fabricated before-and-after photos, fake WhatsApp testimonials, and artificial urgency tactics (e.g., countdown timers and "order now" buttons) [7], as well as pages advertising implausible products such as "smart radiation-proof reading glasses" claiming to sell over 100,000 units per day [8]. The textual patterns of these pages—exaggerated claims, pressure tactics,

and suspicious product descriptions—are amenable to the same transformer-based classification approach.

- **Phishing Content Detection:** Phishing campaigns disseminated through social media messaging share linguistic features with misinformation: impersonation of authority, urgency, and deceptive intent. The framework's containerized deployment stage (Stage 3) is particularly valuable here, as it enables real-time URL analysis that could be extended to phishing detection with minimal architectural changes.

The use of `distilbert-base-multilingual-cased` as the base architecture further enhances transferability, as the same pretrained model supports over 100 languages. Extending the framework to a new language primarily requires the curation of a language-specific corpus (Stage 1) rather than a redesign of the regularization strategy (Stage 2) or the deployment architecture (Stage 3).

5.5. Limitations

Despite its high performance, the model is not infallible. A key limitation is its static nature; it was trained on a closed corpus ending in early 2025. Consequently, the model is susceptible to *concept drift*, where it may fail to detect novel misinformation narratives or evolving linguistic tactics used in future disinformation campaigns. Additionally, the current approach is strictly unimodal (text-only). It fails when the deception is embedded in multimedia elements or when analyzing extremely short texts like tweets or headlines where semantic context is scarce. Furthermore, while the framework is designed to be transferable, the empirical validation presented here is limited to a single domain (Spanish fake news); future work should validate the framework on additional deception domains to strengthen the transferability claim.

5.6. Future Work

Future work will focus on three critical areas to address these limitations and expand the framework's utility. First, the implementation of Continuous Learning (CL) pipelines is envisioned to update the model with fresh data periodically without catastrophic forgetting, mitigating the issue of concept drift. Second, the framework will be empirically validated on additional social media deception domains—specifically, a corpus of investment scams and fraudulent e-commerce pages will be constructed to demonstrate cross-domain transferability. This will involve applying the same three-stage pipeline to these new domains and comparing the resulting models' performance. Third, the expansion of the web application's API to support browser extensions is planned, bringing the detection capability directly to the user's browsing experience and enabling real-time protection against multiple forms of social media deception simultaneously.

Finally, addressing the "black-box" nature of deep learning models remains a priority. Future iterations of the deployed web application will integrate explainability frameworks, such as SHAP (SHapley Additive exPlanations) or LIME, to provide users with calibrated probabilities and visual rationales (e.g., highlighting specific deceptive sentences), thereby enhancing user trust and media literacy. Further research will also incorporate multi-seed training runs and strict calibration metrics (Expected Calibration Error) to provide robust confidence intervals for production environments.

6. Conclusions

This study presents a systematic and transferable framework for fine-tuning pretrained transformer models for domain-specific deceptive content detection on social media. The framework is structured as a three-stage pipeline—corpus unification, systematic regularization, and containerized deployment—designed to be reusable across languages, content domains, and social media platforms.

The framework was validated through a comprehensive case study on Spanish-language fake news detection. A unified corpus of 61,674 articles was constructed from four heterogeneous sources, demonstrating the corpus unification methodology. Through a rigorous experimental process involving over 500 GPU hours, the proposed aggressive regularization strategy for BETO achieved 88.98% accuracy with a generalization gap of only 0.058, significantly outperforming metaheuristic-optimized classical baselines by 16.95 percentage points. The full optimization trajectory—including failed configurations—is documented to provide a reproducible recipe for practitioners.

Beyond the case study, the framework's core technical contribution lies in its transferability. The same three-stage pipeline can be applied to detect other forms of social media deception—including investment scams, fraudulent e-commerce pages, and phishing content—by substituting the training corpus. The deployment of the optimized model into a Dockerized web application demonstrates that the framework extends beyond academic benchmarks to deliver practical tools for digital self-defense. The open-source availability of the entire framework—corpus, model, and application—facilitates further adaptation, paving the way for more resilient detection systems that can protect vulnerable populations against the evolving landscape of social media deception.

Author Contributions: Conceptualization, G.H.A. and J.A.R.-O.; Data Curation, G.H.A. and R.A.M.-G.; Formal Analysis, J.P.C. and Ó.H.A.; Funding Acquisition, J.A.R.-O.; Investigation, G.H.A.; Methodology, G.H.A., J.A.R.-O., R.A.M.-G., J.P.C. and Ó.H.A.; Project Administration, J.A.R.-O.; Resources, G.H.A., J.A.R.-O and Ó.H.A.; Software, G.H.A.; Supervision, J.A.R.-O., R.A.M.-G. and Ó.H.A.; Validation, J.P.C., R.A.M.-G. and Ó.H.A.; Visualization, G.H.A., J.A.R.-O., R.A.M.-G. and J.P.C.; Writing—Original Draft, G.H.A.; Writing—Review and Editing, J.A.R.-O., R.A.M.-G., J.P.C. and Ó.H.A. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The unified corpus and trained models used in this study are publicly available at: https://huggingface.co/datasets/gabrielhuav/Unified-and-Balanced-Spanish-Fake-News-Corpus and https://github.com/gabrielhuav/Spanish-Fake-News-Detection-Training respectively. The source code of the web application is available at: https://github.com/gabrielhuav/Spanish-Fake-News-Detection-Web-App.

Acknowledgments: The authors would like to thank Universidad Autónoma Metropolitana, Unidad Azcapotzalco, and the *Secretaría de Ciencia, Humanidades, Tecnología e Innovación del Gobierno de México (SECIHTI)*, Mexico, for scholarship No. 4013730 (CVU: 1313870), granted to support the development of this research.

Conflicts of Interest: The authors declare no conflicts of interest.

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